

Федеральное государственное бюджетное образовательное учреждение высшего образования «Херсонский государственный педагогический университет»
Институт информационных технологий
Направление подготовки: «09.04.04 Программная инженерия»
Специализация «Разработка программно-информационных систем»
Форма обучения: очная
Курс: 1, Магистратура, группа 12-25РПм
Херсонская область - 2025

Дисциплина:
«Иностранный язык для специальных целей»

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ПРАКТИЧЕСКАЯ РАБОТА №5 Exercise 15. Strategies for reading academic texts: before reading

I. ЦЕЛЬ И ЗАДАЧИ РАБОТЫ

1.1. Цель работы

Освоить стратегии эффективного чтения академических текстов: подготовка к чтению, прогнозирование содержания и формулировка вопросов.

1.2. Задачи работы

- Развивать навыки понимания академических текстов.
- Научиться использовать предиктивное чтение: заголовки, подзаголовки, иллюстрации.
- Развивать умение выделять ключевые слова и формулировать вопросы по теме.
- Совмещать чтение на английском с переводом и пониманием на русском языке.

II. ЗАДАНИЯ И ВЫПОЛНЕНИЕ

2.1. Think about your reasons for reading the text

- a. I am interested because it is about my subject. Answer: I read the text to gain more knowledge about scientific words. Перевод: Я читаю текст, чтобы углубить свои знания о научной терминологии.
- b. I want background information and detailed information. Answer: I want to understand the history, structure, and meaning of scientific words. Перевод: Я хочу понять историю, структуру и значение научных слов.
- c. I want to know what the writer's views are. Answer: The author emphasizes the importance of precision and international understanding in scientific language. Перевод: Автор подчеркивает важность точности и международного понимания в научной терминологии.
- d. I am going to have a discussion. Answer: I can use the information to discuss the formation and meaning of scientific words. Перевод: Я могу использовать информацию для обсуждения формирования и значения научных терминов.
- e. I am going to write an essay on this subject later. Answer: I will use the strategies to extract key points and examples for my essay. Перевод: Я буду использовать стратегии для выделения ключевых моментов и примеров для моего эссе.

2.2. Look at the title, headline, sub-headings, and illustrations

- Task: Predict the topic. Answer: The text is about scientific words, their structure, origin, and meaning. Перевод: Тема текста — научные слова, их структура, происхождение и значение.

2.3. Think about what you already know on this topic

- Task: Recall prior knowledge. Answer: Scientific words often come from Latin or Greek, have precise meanings, and are used internationally. Перевод: Научные термины часто происходят из латинского или греческого языка, имеют точные значения и используются международно.

2.4. Write down what you would like to find out from the text

- Task: Formulate questions. Answer: - How are scientific words formed? - What are the different sources of scientific words? - Why is precision important in scientific language? - How do classical languages influence scientific terms? Перевод: - Как формируются научные слова? - Какие существуют источники научной терминологии? - Почему точность важна в научном языке? - Как классические языки влияют на научные термины?

2.5. Make a note of words or phrases connected with the topic

- Task: Identify key vocabulary. Answer: - Vocabulary - Scientific words - Technical terms - Precision - Latin, Greek roots - Structure, meaning - Invention of terms - International usage - Classical languages - Communication
Перевод: - Словарный запас - Научные слова - Технические термины - Точность - Латинские и греческие корни - Структура, значение - Изобретение терминов - Международное использование - Классические языки - Общение

III. АНАЛИЗ ТЕКСТА

3.1. The Purpose and Nature of Scientific Words

- Task: Summarize main ideas. Answer: Scientific words provide precise names for substances, processes, and abstractions. They allow scientists to communicate clearly and avoid ambiguity. Перевод: Научные слова дают точные названия веществ, процессов и абстракций. Они позволяют ученым общаться ясно и избегать двусмысленности.

3.2. Sources of Scientific Words

- Task: Identify sources. Answer: 1. Words taken from ordinary English vocabulary and given special meanings. 2. Words borrowed from other languages with little or no change. 3. Words invented by scientists, often using classical roots. Перевод: 1. Слова из обычного английского, получившие специальные значения. 2. Заимствованные слова из других языков без изменений или с минимальными изменениями. 3. Слова, изобретенные учеными, часто на основе классических корней.

3.3. Importance of Classical Languages

- Task: Explain why classical languages are used. Answer: Latin and Greek provide roots and elements distinct from everyday speech, helping create precise and internationally understood scientific terms. Перевод: Латинский и греческий языки дают корни и элементы, отличающиеся от повседневной речи, что помогает создавать точные и международно понятные научные термины.

IV. ВЫВОДЫ

4.1. Summary

- Using pre-reading strategies helps to focus, predict content, and formulate questions. Understanding the origin, structure, and precision of scientific words is crucial for effective reading and academic communication. Перевод: Использование стратегий подготовки к чтению помогает сосредоточиться, предсказать содержание и сформулировать вопросы. Понимание происхождения, структуры и точности научных слов важно для эффективного чтения и академического общения.

Exercise 15. Strategies for reading academic texts: before reading.

1. Think about your reasons for reading the text:
 - you are interested because it is about your subject, or it is related to your subject
 - you want background information, or detailed information
 - you want to know what the writer's views are
 - you are going to have a discussion
 - you are going to write an essay on this subject later.

Each reason will influence the way you read e.g., quickly or slowly, looking for fact or opinion.

2. Look at the title, headline, any sub-headings, photos or illustrations. Use these elements to predict what the text will be about — the topic.

3. Think about what you already know on this topic.

4. Write down what you would like to find out from the text. You could write actual questions you would like answers to.

5. Make a note of words or phrases connected with the topic that you may find in the text.

Use the strategies above on the following text.

Scientific Words: Their Structure and Meaning

The Purpose and Nature of Scientific Words

The development of an appropriate vocabulary is essential to the development of any subject. Words are the elements of language; language is the vehicle of ideas. By silent language thoughts are developed in the mind, and by written or spoken language thoughts are communicated to others.

It is obvious that a scientist must have names by which to identify and refer to the various chemical substances, minerals, plants, animals, structural units, instruments, etc., with which he deals. He must have suitable adjectives for describing these things and suitable verbs for defining their behaviour.

He also needs suitable names by which to identify the various abstractions with which he deals — processes, states, qualities, relationships, and so on. Thus, after Faraday had investigated the passage of electric currents through different solutions and noted the resulting liberation of chemical substances, the term *electrolysis* was invented. This one word was a kind of

shorthand symbol for the process; it 'pinned down' the process and conveniently embraced its many aspects. From then on it was possible to think about the process and to talk about it to others. Similarly, the single term *symbiosis* conveniently summarises a biological state; *diathermancy* identifies a physical quality.

Many scientific words are of this kind. Without the name (or technical term) a concept remains vague and ill-defined; the scientist is hindered in his mental processes, in his recording of what he thinks and does, and in his communication with others.

In his communication a scientist is mainly concerned with the exact and logical expression of that which he wishes to pass on to another. His purpose is to inform (as clearly as possible), not to excite emotion. It follows that each of his words must have a precise meaning, and one meaning only, so that there is no risk of confusion or ambiguity. Of course he must know himself what his words mean and he must assume that the person with whom he communicates attaches the same meanings to them. (If he is communicating with a person who is unlikely to understand his specialised terms, he must take care not to use them, even if that may mean some loss of precision or elegance. A number of 'popular' science talks fail because the speaker, often an expert scientist, thoughtlessly uses words which the ordinary person does not understand.)

The meanings of many ordinary words of our language are not single and precise. Although the original, basic meanings may be clear, the words have acquired a range of meanings over the years. Thus the familiar word *fair* has somewhat different meanings when used to describe the weather, a person's hair, an action or decision, or a boy's performance at school; some words (e.g., '*rude*') suffer a significant change in meaning. Hence a scientist avoids the ordinary words of the language; he prefers his own words. These words can then be rigorously defined and given the necessary precision of meaning.

The use of words which are 'set apart' from everyday life also enables the scientist to avoid evoking irrelevant and distorting associations. Some ordinary words convey more than their literal meanings; they evoke further images, emotions and reactions on the part of the hearer or reader. (Thus *red*, basically a word denoting a certain colour, may conjure up thoughts and feelings relating to danger, to blood, or to a particular political outlook.) The specialised words of science, if used in their proper contexts, are largely free from distorting associations. It is interesting to note that when a scientific term, originally well-defined, becomes a word of ordinary speech, it usually suffers a widening of meaning and acquires a number of associations. Thus criticism (as well as sulphuric acid) may be *vitriolic*, a man may be *electrified* into action, and people may claim to be *allergic* to all sorts of things and conditions. The word *atomic*, whose meaning is quite clear to the scientist, may conjure up in the public mind a picture of wide spread destruction or of unlimited power.

In addition to precision of meaning and freedom from associations, most scientific words have a third quality: by their form and structure they reveal something of their meanings. Many scientific words are logically built up from simpler word-elements (usually of Greek or Latin origin) and the general meaning of the whole can be inferred from an understanding of the parts. Some terms, in fact, are self-explanatory if the Latin and Greek roots are known; they have only to be 'translated' for their meanings to become apparent.

Thus a *quadrilateral* is clearly a four-sided figure, *entomology* is the study of insects, *gastrectomy* is the cutting out of the stomach (or part of it). In the case of a large number of words the full or precise meaning may not be directly disclosed but the general meaning is apparent and the word is seen to 'make sense'. Thus *cyanosis* indicates a state (possibly a morbid state) of blueness; it is a sensible word to use to denote the blue condition of the skin which results from insufficient oxygen in the blood. A *xerophyte* (literally 'a dry plant') is one which is adapted for living in very dry conditions; a *hydrophyte* is one which lives on the surface of, or submerged in, water. A *polymer* consists of 'many parts'; the term is an appropriate one for a giant molecule which is built up from a large number of simple units.

In a similar way, many chemical names are essentially descriptions of the compounds which they denote. Thus whereas the common name *aniline* for a certain oil discloses nothing about the nature of the compound (except, perhaps, that it is vaguely related to indigo), the chemical name *aminobenzene* immediately indicates the molecular composition and structure.

Scientific language, to be efficient, must be universally intelligible. The classical languages, Latin and Greek, are so fundamental to the civilised world that words constructed from elements of these languages are readily understood the world over. (Even if scientists know little of the classical languages, they can easily learn to 'translate' the scientific terms which they may meet.) Most scientific terms are effectively international.

Sources of Scientific Words

Scientific words in English may conveniently be divided, from the standpoint of their origins, into three groups:

- a. those taken from the ordinary English vocabulary;
- b. those taken virtually unchanged from another language;
- c. those which have been invented.

The third group is by far the largest.

Just as the cricketer has taken certain everyday words, such as *run*, *over*, *maiden*, from the general English vocabulary and given them specialised meanings within the context of his game, so the scientist has occasionally taken ordinary English words and endowed them with specialised meanings. *Energy*, *work*, *power*, *salt*, *base*, *fruit* are examples of such words. They are unsatisfactory as scientific terms because they lack the essential qualities which we have described. Although the scientist may give them precise meanings, they are liable to be interpreted more loosely (or even differently) by the non-scientist.

They are not free from irrelevant associations; they reveal little of their meanings by their forms; and usually, they are not understood outside the English-speaking countries. There are not many words of this kind but, unfortunately, most of them stand for concepts of fundamental importance.

The English language contains a number of words which have been taken from another language with little or no change of spelling. Amongst them are *caf*, *morgue*, *souvenir*, *trek*, *marmalade* and *agenda*. Practically all the scientific words of this kind have been taken from Latin or Greek. As examples of Latin words we may note *axis*, *fulcrum*, *larva*, *radius*, *locus*, *nimbus*, *cortex*. Many parts of the human body, e.g., *cerebrum*, *pelvis*, *cornea*, have Latin names. There are fewer unaltered Greek words — *thorax*, *stigma*, *iris*, *helix* are examples — but it should be noted that many terms adopted in Latin form, e.g., *trachea*, *bronchus*, *phylum*, were themselves based on Greek. Many of the Greek or Latin terms have retained their original meanings but in some cases the meanings have been restricted and rendered more precise.

The largest group of scientific words is those which have been invented. The advance of science during the last few centuries has been so rapid and so extensive that no language has been capable of providing, ready-made, all the words which were required. Further, the classical languages do not contain words appropriate to modern discoveries, inventions and concepts. (There is no Latin word, for example, for photography!) Hence the scientist has had to invent new words for his own purposes.

It is very rare for a scientist to make up a word 'out of his head'; the term *ester* for a compound formed by the interaction of an alcohol and an organic acid was perhaps such an invention. A small but interesting group of terms comprises those based on proper names. In the naming of the chemical elements recourse has been made to the names of places (as in *polonium*, *ytterbium*), of gods and goddesses (as in *thorium*, *vanadium*), of planets and asteroids (as in *uranium*, *cerium*), and of scientists themselves (as in *curium*, *gadolinium*). Scientists' names have also been used to provide the names of units (e.g., *watt*, *volt*, *gauss*, *joule*) and hence the names of measuring instruments (e.g., *voltmeter*). Among the other terms based on the names of scientists are *daltonism*, *nicotine*, *bakelite* and *mendelism*. A number of plants, e.g., *fuchsia*, *dahlia* are named after botanists.

In his task of inventing new terms, however, the scientist has usually turned to the classical languages for his raw material. He has taken 'bits and pieces' — roots, prefixes, suffixes — from these languages and joined them together to form the terms he needed. Thus, when he needed a general name for animals such as snails and slugs which apparently walk on their stomachs, he took the Greek roots *gast(e)ro-* (stomach) and *-pod* (foot) and formed the new word *gastropod*. When he wanted a word to describe a speed greater than that of sound he took the Latin prefix *super-* (above, beyond) and the Latin root *son-* (sound) and coined the adjective *supersonic*. Thousands of scientific words have been built up from classical word-elements in this way.

It may be asked why the scientist should have turned to the classical languages for the words and word-elements which he needed. By turning to a language other than his own he was certainly able to find words and elements which were distinct from those of ordinary speech but he turned to the classical languages for an important historical reason. The fifteenth and sixteenth centuries witnessed that great revival of classical learning which is commonly called the Renaissance. Latin was regarded as the universal language of scholarship; it was the 'perfect' language of philosophy, theology and science. This classical tradition persisted into the seventeenth century — both Harvey and Newton wrote their great works in Latin — and it was not until towards the end of that century that English was fully accepted as an adequate and suitable language for a scholarly exposition of science.

During this period many Latin words were taken into the scientific vocabulary and many new words were constructed (chiefly in the form of Latin words) from classical elements. The tradition of using the classical languages as a source of scientific words remains.

Greek was not used in the same way as a medium of expression but it was held in respect as the language of the people who at one time led the world in art, science and philosophy. Moreover, it provided a particularly suitable basis for scientific language. It had been developed by a long line of philosophers as a medium for accurate expression and its elements were such that derivatives and compounds were readily formed. The scientists therefore mainly went to the Greek for the new terms which they needed (though, as has been pointed out, the terms were at one time often framed in Latin form). Greek is still the source of most of the new terms of science and more than half of the words of the great vocabulary of science are ultimately of Greek origin.